



METABOLIC & GLP-1

5-Amino-1MQ

5-Amino-1MQ – 50 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	5-Amino-1-methylquinolinium iodide
SYNONYMS / ALIASES	5-Amino-1MQ; 5A1MQ; selective NNMT inhibitor
CAS NUMBER	Not assigned in standard chemical registries (research compound; see published references for structural identification)
MOLECULAR FORMULA	C10H11IN2 (iodide salt); cation: C10H11N2+
MOLECULAR WEIGHT	Approximately 286.11 g/mol (iodide salt)
AMINO ACID SEQUENCE	Not applicable - small molecule, not a peptide
SOURCE	Synthetic (organic chemistry; quaternary quinolinium salt)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized or crystalline solid; color may range from off-white to pale yellow
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	50 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C in aqueous solution; stability is solvent-dependent
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only); soluble in DMSO for in vitro work
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Quaternary ammonium salts are generally water-soluble and chemically stable. Protect from prolonged light exposure.

STRUCTURAL NOTES

5-Amino-1MQ is a small-molecule quaternary quinolinium compound, not a peptide. It is structurally a methylquinolinium scaffold bearing a primary amine at the 5-position of the quinoline ring and a methyl group on the ring nitrogen, present as the iodide salt. It is classified in the research literature as a selective small-molecule inhibitor of nicotinamide N-methyltransferase (NNMT).

SELECTED PEER-REVIEWED REFERENCES

1. Neelakantan H. et al. - *Selective and membrane-permeable small molecule inhibitors of nicotinamide N-methyltransferase reverse high fat diet-induced obesity in mice* - Biochemical Pharmacology, 2018 - DOI: 10.1016/j.bcp.2018.01.040
2. Kannt A. et al. - *A small molecule inhibitor of Nicotinamide N-methyltransferase for the treatment of metabolic disorders* - Scientific Reports, 2018 - DOI: 10.1038/s41598-018-22081-7
3. Neelakantan H. et al. - *Structure-activity relationship for small molecule inhibitors of nicotinamide N-methyltransferase* - Journal of Medicinal Chemistry, 2017 - DOI: 10.1021/acs.jmedchem.7b00389

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.